

AKROMID®

B3 ICF 30 black (5119)

PA6 CF30

AKROMID® B3 ICF 30 black (5119) is a 30% carbon fibre reinforced polyamide 6 with high flexural strength and good sliding properties

Properties

Modulus	Strength	Impact
23.000 MPa	200 MPa	55 kJ/m ²

Mechanical Properties

Tensile modulus ISO 527-2	1 mm/min d.a.m.	23000 MPa
	1 mm/min conditioned	11000 MPa
Tensile stress at break ISO 527-2	5 mm/min d.a.m.	200 MPa
	5 mm/min conditioned	125 MPa
Tensile strain at break ISO 527-2	5 mm/min d.a.m.	2 %
	5 mm/min conditioned	4 %
Flexural modulus ISO 178	2 mm/min d.a.m.	18000 MPa
	2 mm/min conditioned	11300 MPa
Flexural strength ISO 178	2 mm/min d.a.m.	290 MPa
	2 mm/min conditioned	200 MPa
Flexural strain at break ISO 178	2 mm/min d.a.m.	2 %
	2 mm/min conditioned	3,5 %
Charpy impact strength ISO 179-1/1eU	23°C d.a.m.	55 kJ/m ²
	23°C conditioned	70 kJ/m ²
	-30°C d.a.m.	50 kJ/m ²
Charpy notched impact strength ISO 179-1/1eA	23°C d.a.m.	10 kJ/m ²
	23°C conditioned	15 kJ/m ²
	-30°C d.a.m.	7 kJ/m ²

Disclaimer

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Thermal Properties

Temperature of deflection under load HDT/A	1,8 MPa	210 °C
ISO 75		
Melting temperature	DSC, 10K/min	220 °C
ISO 11357-3		
Coefficient of linear thermal expansion	23°C to 80°C parallel	0,6 10⁻⁴/K
ISO 11359-1/2	23°C to 80°C transverse	0,93 10⁻⁴/K

Flammability

Burning rate	1,6 mm Wall thickness	HB Class
UL 94		
Burning rate (<100 mm/min)	> 1 mm Thickness	+
FMVSS 302		

General properties

Density	23°C	1,28 g/cm³
ISO 1183		
Humidity absorption	70°C, 62% r.H.	2,1 %
ISO 1110		
Molding shrinkage	flow	0,1 - 0,3 %
ISO 294-4	transverse	0,4 - 0,6 %

Electrical Properties

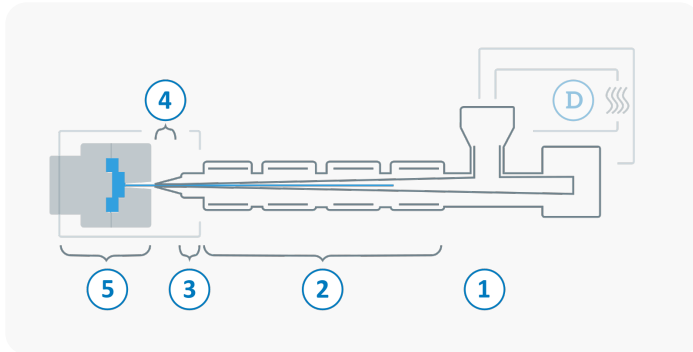
Surface resistivity	d.a.m.	10³ Ω
DIN EN 62631-3-2	conditioned	10³ Ω

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Processing

The listed values are recommendations. Higher values should be used for higher glass loadings. We recommend only de-humidifying or vacuum dryers. Extensive drying can cause filling problem and surface defects.



D	Drying time	0 - 4 h
	Drying temperature ($\tau \leq -30^{\circ}\text{C}$)	80 °C
	Processing moisture	0,02 - 0,1 %
1	Feed section	60 - 80 °C
2	Temperature Zone 1 - Zone 4	240 - 300 °C
3	Nozzle temperature	270 - 300 °C
4	Melt temperature	270 - 300 °C
5	Mold temperature	80 - 130 °C
→	Holding pressure, spec.	300 - 800 bar
←	Back pressure, spec.	50 - 150 bar
	Injection speed	medium to high
	Screw speed	8 - 15 m/min

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